# ARJUNA JEE 2027

Lecture -06 | Mathematics

# PERMUTATIONS AND COMBINATIONS

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# Topics *to be covered*

A **Gap & Tie Method**

B **Arrangement of things not all Different**

C **Question Practice**

D





# LAST CLASS RECAP

# In '*n*' sided polygon the triangles are formed by joining the vertices of polygon then:

(a) Number of triangles having one side common with the sides of polygon = $n(n-4)$

(b) Number of triangles having two side common with the sides of polygon = $n$.

(c) Number of triangles having no side common with polygon = ${}^{n}C_{3} - (n^2 - 3n)$

(d) Number of diagonals in 'n' – sided polygon = ${}^{n}C_{2} - n$

# H.W. Discussion

Total 5 different pair of shoes are there, you need to select only 4 shoes such that

(i) No restriction $= {}^{10}C_4 = 210$.

(ii) Exactly 1 pair of should be formed

(iii) Exactly 2 pair of should be formed $= {}^{5}C_2 = 10$.

(iv) No pair should form

| |
|---|
| $L_1R_1$ ✓ |
| $L_2R_2$ |
| $L_3R_3$ |
| $L_4R_4$ ✓ |
| $L_5R_5$ |

Ex:- $L_1R_1\ L_2L_3$

$L_2R_2\ R_3L_1$

⋮

(ii) ${}^{5}C_1 \times \left( \overset{LL}{{}^{4}C_2} + \overset{RR}{{}^{4}C_2} + \overset{LR}{{}^{4}C_1 \cdot {}^{3}C_1} \right)$

M-I $\rightarrow 5\,(6+6+12) = 5 \times 24 = 120$.

OR

M-II ${}^{5}C_1 \times \left( {}^{8}C_2 - \underbrace{4}_{\text{when pair is selected}} \right) = 5 \times (28-4) = 5 \times 24 = 120$.

M-III ${}^{5}C_1 \times {}^{4}C_2 \times {}^{2}C_1 \times {}^{2}C_1$

$= 5 \times 6 \times 2 \times 2$

(iv) No pair

$^{10}C_4$ → No pair
$^{10}C_4$ → ex. 1 pair ✓
$^{10}C_4$ → ex. 2 pair ✓

**M-I :-**

$$\text{fnf} = {}^{10}C_4 - (120 + 10)$$
$$= 210 - (130)$$
$$= 80.$$

**M-II :-**

$L_1R_1$ $L_2R_2$ $L_3R_3$ $L_4R_4$ $L_5R_5$

$${}^5C_4 \times {}^2C_1 \times {}^2C_1 \times {}^2C_1 \times {}^2C_1$$
$$5 \times 16 = 80 \checkmark$$

## Question [IIT-JEE-2005]

65%



There is a rectangular sheet of dimension $(2m-1) \times (2n-1)$, (where $m > 0$, $n > 0$). It has been divided into square of unit area by drawing line perpendicular to the sides. Find the number of rectangles having sides of odd unit length.

A $(m+n+1)^2$

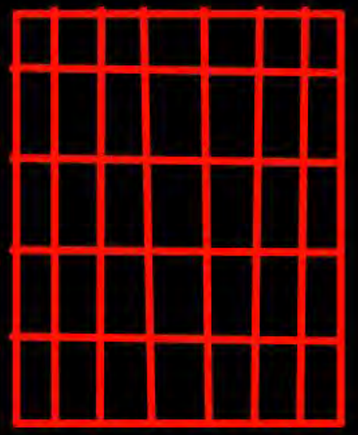
B $mn(m+1)(n+1)$

C $4^{m+n-2}$

D $m^2n^2$

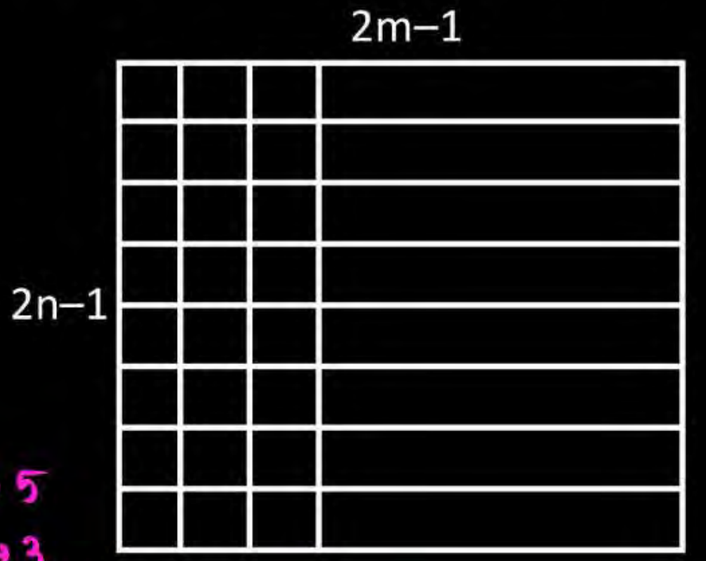
$m=3, \quad n=4$

Rectangular sheet → $5 \times 7$




Odd dimensions

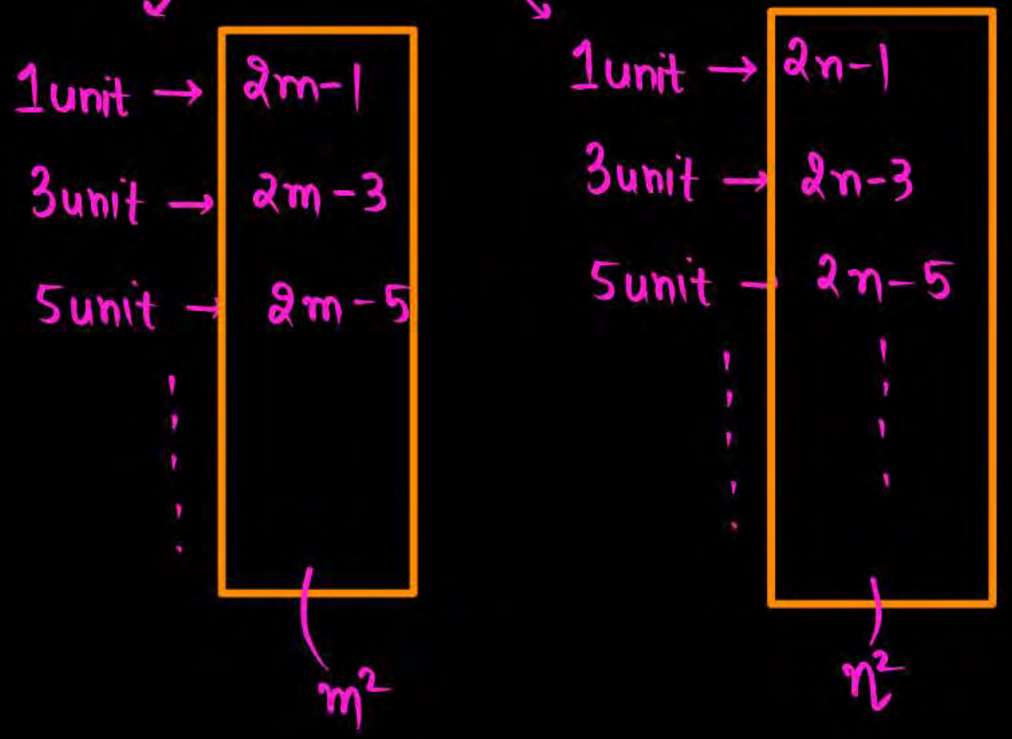
| | |
|---|---|
| 1 unit → 7 | 1 unit → 5 |
| 3 unit → 5 | 3 unit → 3 |
| 5 unit → 3 | 5 unit → 1 |
| 7 unit → 1 | |
| 16 | 9 |




$^{16}C_1 \times {}^{9}C_1 = 16 \times 9$

$= n^2 \cdot m^2$

$(2m-1) \times (2n-1)$



# TIE & GAP METHOD

## Question

(i) Arrange $a, b, c, d, e$ & $f$ in a line such that $a, b$ & $c$ are always together.

ex:- abcdef

dfcbae

- Tie Method

abc, d, e, f → 4 units.

$4! \times 3!$

Tie करने के बाद internal arrange.



# TIE & GAP METHOD



## Question

(ii) Arrange $(a, b, c, d, e \text{ \& } f)$ in a line such that no two of $a, b$ & $c$ are together.

ex:- b d a e f c ✓

a f c d b e ✓

⋮

Gap Method : (jinko alag-alag rakhna hai unko GAP में बैठाओ)

_ d _ e _ f _

a d e b f c

${}^4C_3 \times 3! \times 3!$

- ${}^4C_3$ → selec$^n$. of gap.
- $3!$ → arr. on GAP.
- $3!$ → arr. of d, e, f.

# TIE & GAP METHOD

## Question

(iii) Arrange $(a, b, c, d, e \text{ \& } f)$ in a line such that $a, b$ & $c$ are not all together.

Bijection

abdecf

adebfc

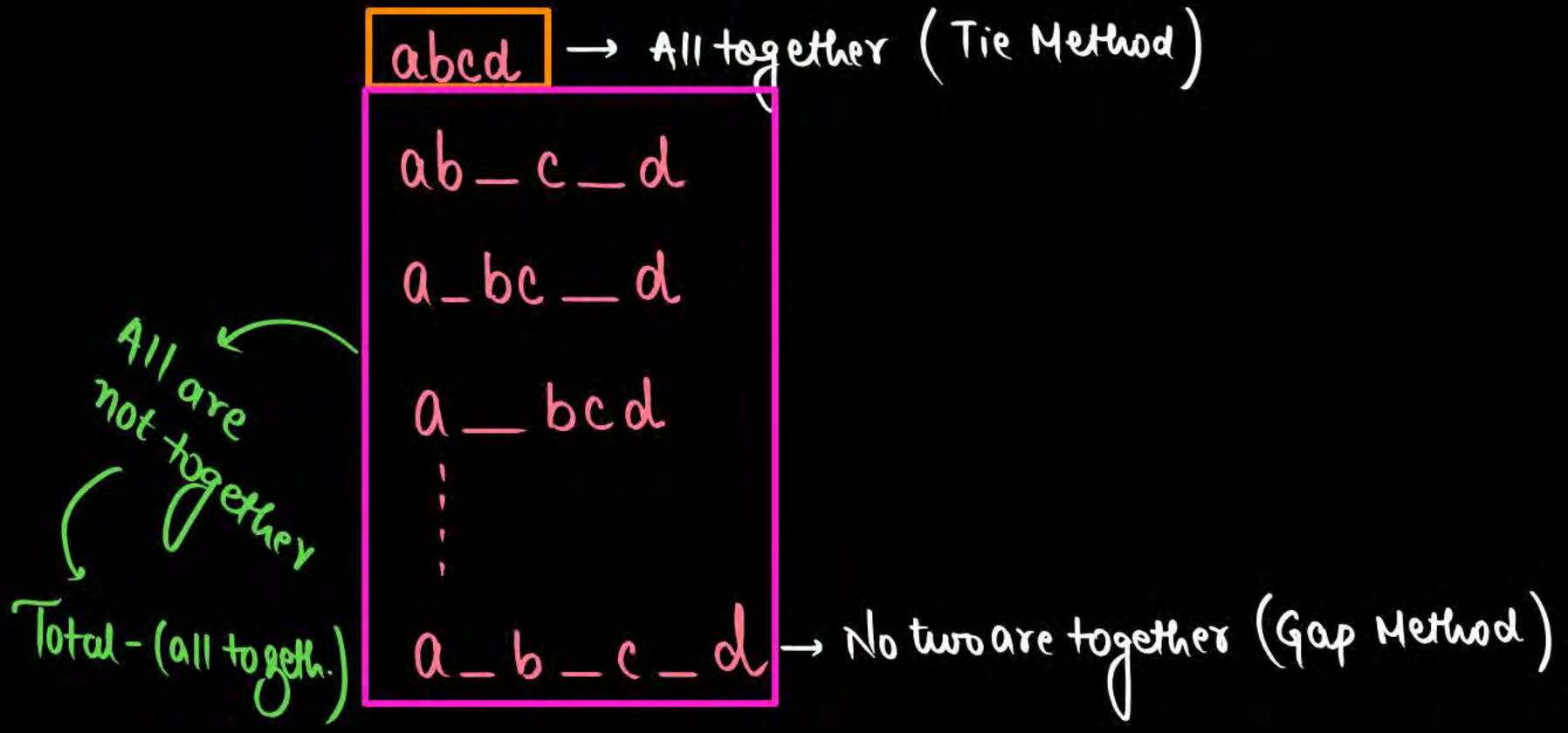
aebcdf

⋮

fnf = Total − (all together)

= $6!_0 - (4!_0 \times 3!_0)$

# CONSIDER FOLLOWING CASES



# SACHIN

all vowels are not together

$\underline{\underline{L}}\ 6!_0 - (5!_0 \times 2!_0)$

$6.5!_0 - 5!_0 \times 2!_0$

$5!_0\,(6-2)$

$4 \times 120 = 480.$

no two vowels are together.

S, C, H, N  A, I

$\underline{\underline{\checkmark}}\,S_\,C\,\underline{\underline{\checkmark}}\,H_\,N_$

$4!_0 \times {}^5C_2 \times 2!_0 = \underline{\underline{480}}.$

$24 \times 10 \times 2$

vowels are together.

S, C, H, N, (A, I)

$5!_0 \times 2!_0$

Blue बादल (KRPNCC)

जब सबको एक साथ रखना हो (All are together)

use: TIE METHOD

Blue बादल (KRPNCC)

(कोई भी दो एक साथ ना हो) (No two are together)

use: GAP METHOD

ex:-

5 Boys, 4 Girls → arrange in a line. $B_1B_2B_3B_4B_5$ $G_1G_2G_3G_4$

① all boys are together. → $(B_1B_2B_3B_4B_5)$, $G_1, G_2, G_3, G_4$ → $5!_0 \times 5!_0$

② all girls are together. → $B_1, B_2, B_3, B_4, B_5$, $(G_1G_2G_3G_4)$ → $6!_0 \times 4!_0$

③ no two girls are together.

$$-B_1-B_2-B_3-B_4-B_5-$$

$$5!_0 \times {}^6C_4 \times 4!_0$$

④ no two boys are together.

$$-G_1-G_2-G_3-G_4-$$

$$4!_0 \times {}^5C_5 \times 5!_0$$

## Bumper Question



**4 boys & 4 girls are to be seated in a line find number of ways:**

(i) If all boys are together & all girls are together

(ii-a) They can be seated so that No two girls are together

(ii-b) If not all the girls are together **OR**
If at least one girl is separated from rest of girls.

$B_1\ B_2\ B_3\ B_4$ $G_1\ G_2\ G_3\ G_4$

(i) $B_1\ B_2\ B_3\ B_4$ $G_1\ G_2\ G_3\ G_4$

$(i) = 2!_0 \times 4!_0 \times 4!_0$

(ii) $-B_1-B_2-B_3-B_4-$

$4!_0 \times {}^5C_4 \times 4!_0$

(iii) Bijection

Total − (all girls are together)

$8!_0 - (5!_0 \times 4!_0)$

## Bumper Question



**4 boys & 4 girls are to be seated in a line find number of ways:**

(iii) Boys and girls are alternate

(iv) If there are 4 married couples then number of ways in which they can be seated so that each couple is together.

B G B G B G B G → $4! \times 4!$

G B G B G B G B → $4! \times 4!$

(iii) = $4! \times 4! \times 2$

(iv)



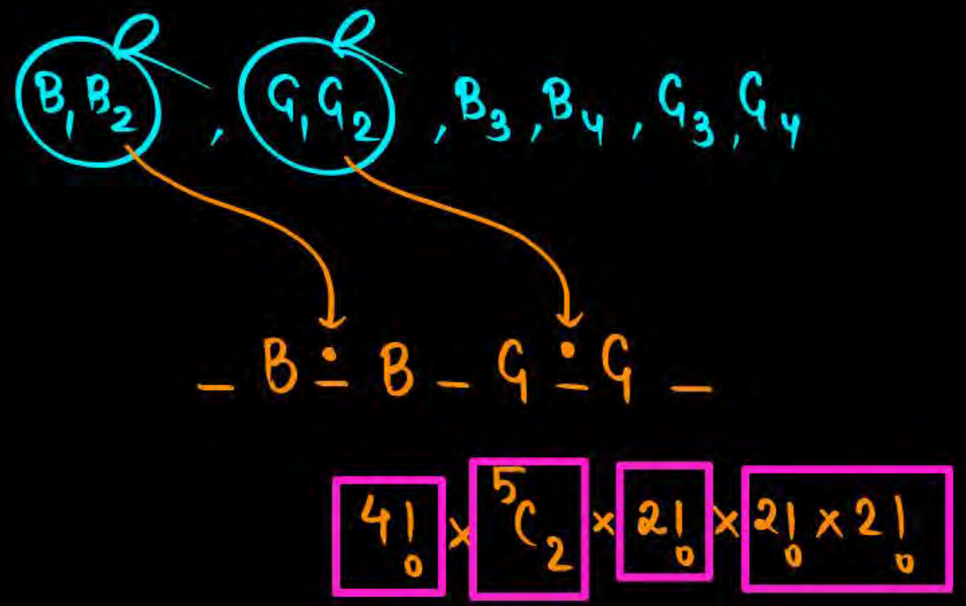

$4! \times 2! \times 2! \times 2! \times 2!$

## Bumper Question



**4 boys & 4 girls are to be seated in a line find number of ways**

(v) $B_1$ & $B_2$ are together, $G_1$ & $G_2$ are together but they all should not be together.



## Bumper Question

**4 boys & 4 girls are to be seated in a line find number of ways**

(vi) Between $B_1$ & $G_1$ there are exactly 4 persons.

(vii) Between $B_1$ & $G_1$ there at least 4 persons.

$B_2 B_3 B_4 G_2 G_3 G_4$

(vi)

$B_1 - - - - G_1 - -$ → $2! \times 6!$

$- B_1 - - - - G_1 -$ → $2! \times 6!$

$- - B_1 - - - - G_1$ → $2! \times 6!$

$3 . 2! \times 6!$

$3! \times 6!$

(vii)

(I) $B_1, G_1$ ke beech ex. 4 persons → $3! \times 6!$

(II) " " " " 5 " → $2 . 2! . 6!$

(III) " " " " 6 persons:

$B_1 - - - - - G_1 -$ → $2! \times 6!$

$- B_1 - - - - - G_1$ → $2! \times 6!$

$B_1 - - - - - - G_1$

$2! \times 6!$

## Bumper Question

"Re-attempt"

**4 boys & 4 girls are to be seated in a line find number of ways**

**(viii) Boy and girl are alternate and a particular boy & girl are never adjacent to each other in any arrangement.**

already fixed $B_1, G_1$.

ex:- $B_2\ G_2\ B_1\ G_3\ B_3\ G_1\ B_4\ G_4$

Bijection

Total alternate wale cases → $4! \cdot 4! \cdot 2$

Total alternate wale cases → $B_1G_1$ together

Total alternate wale cases → Not together. (Required)

(VIII) M-1

Bijection

Total alternate wale cases → $4! \cdot 4! \cdot 2$

Total alternate wale cases → $B_1G_1$ together

Total alternate wale cases → Not together. (Required)

$B_1G_1$

$_B_G_B_G_B_G_$

$^7C_1 \times 1 \times (3! \times 3! \times 2!)$

$fnf = 4! \cdot 4! \cdot 2 - (7 \times 3! \times 3! \times 2)$

$= 24 \times 24 \times 2 - (7 \times 6 \times 6 \times 2)$

$= 12 \times 6\,(2 \times 4 \times 2 - 7)$

$= 72 \times 9 = 648$

$B_1\ B_2\ B_3\ B_4\ \ G_1 G_2 G_3 G_4$.

$\text{Ans} = 648$

**M-2**

B $G_1$ B G B $G_1$ G B $G_2$

G B G B G B $G_2$ B

$216 = {}^3C_1 \times {}^2C_1 \times 3!_0 \times 3!_0$.

$B_1$ G B $G_1$ B G B $G_2$

$108 = 1 \times {}^3C_1 \times 3!_0 \times 3!_0$

324

324

- 10 shoes → ${}^{10}C_4 = 210$



$10 \times 9 \times 8 \times 7 = 5040 = {}^{10}C_4 \times 4!$

arrangement

6 diff. Things

Select any 3.

${}^{6}C_3 = 20$ ✓

${}^{6}C_1 \times {}^{5}C_2 = 60$ ✗

${}^{6}C_1 \times {}^{5}C_1 \times {}^{4}C_1 = 6 \times 5 \times 4$

$= 120$ ✗



# ARRANGEMENT WHEN ALL THINGS ARE NOT DIFFERENT



*# Number of arrangements of A,B,C,D,E (taken all at a time) =* $5!$

*# Number of arrangements of A,A,A,B,C (taken all at a time) =* $\frac{5!}{3!}$

(3-alike)

*# Number of arrangements of A,A,A,B,B,C,C (taken all at a time) =* $\frac{7!}{3!2!2!}$

*# Number of arrangements of all the letters of the word MISSISSIPPI =* $\frac{11!}{4!4!2!}$

S → 4
I → 4
P → 2
M → 1

A, B, C

$3! = 6$

ABC
ACB
BCA
BAC
CAB
CBA

A, A, A

AAA → 1

A, A, B

AAB
ABA
BAA

} 3 ways

A, A, A, B, C

÷ — ÷ ÷ —

$^{5}C_{3} \times 1 \times 2!_{0} = \frac{5!_{0}}{3!_{0}\cancel{2!_{0}}} \times \cancel{2!_{0}}$

AAABBCC

B C A B A C A

$$^{7}C_{3} \times {}^{4}C_{2} \times 1$$

$$\frac{7!}{3!\,4!} \times \frac{4!}{2!\,2!} = \frac{7!}{3!\,2!\,2!}$$

## Question

Number of words with or without meaning that can be formed using all the letters of the word ASSASSINATION that:

(i) Start & end with 'S'.

(ii) ends with vowel

(iii) [JEE MAINS-2023] Vowels occurs together

Total = 13

4 → S

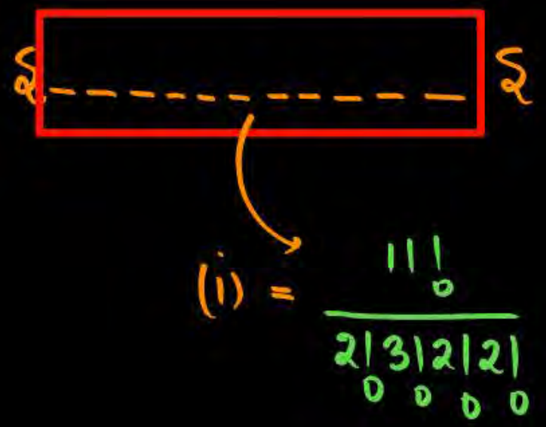
3 → A ✓

2 → I ✓

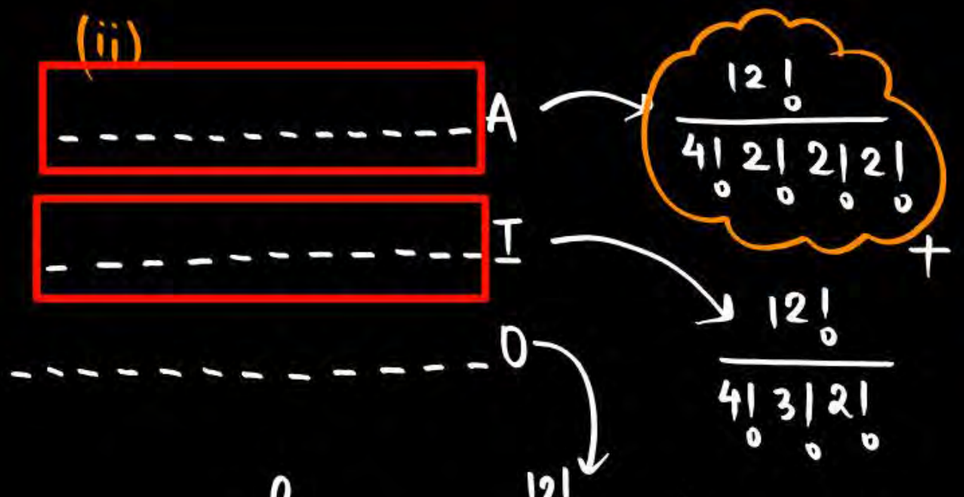
2 → N

1 → T

1 → O ✓




$(i) = \frac{11!}{2!3!2!2!}$




$\frac{12!}{4!2!2!2!} + \frac{12!}{4!3!2!}$

$fnf = \frac{12!}{4!3!2!2!} +$

(iii) AAAIIO, SSSSS, N, N, T

$fnf = \left(\frac{8!}{4!2!}\right) \times \left(\frac{6!}{3!2!}\right)$

## Question [IIT-JEE-2009]

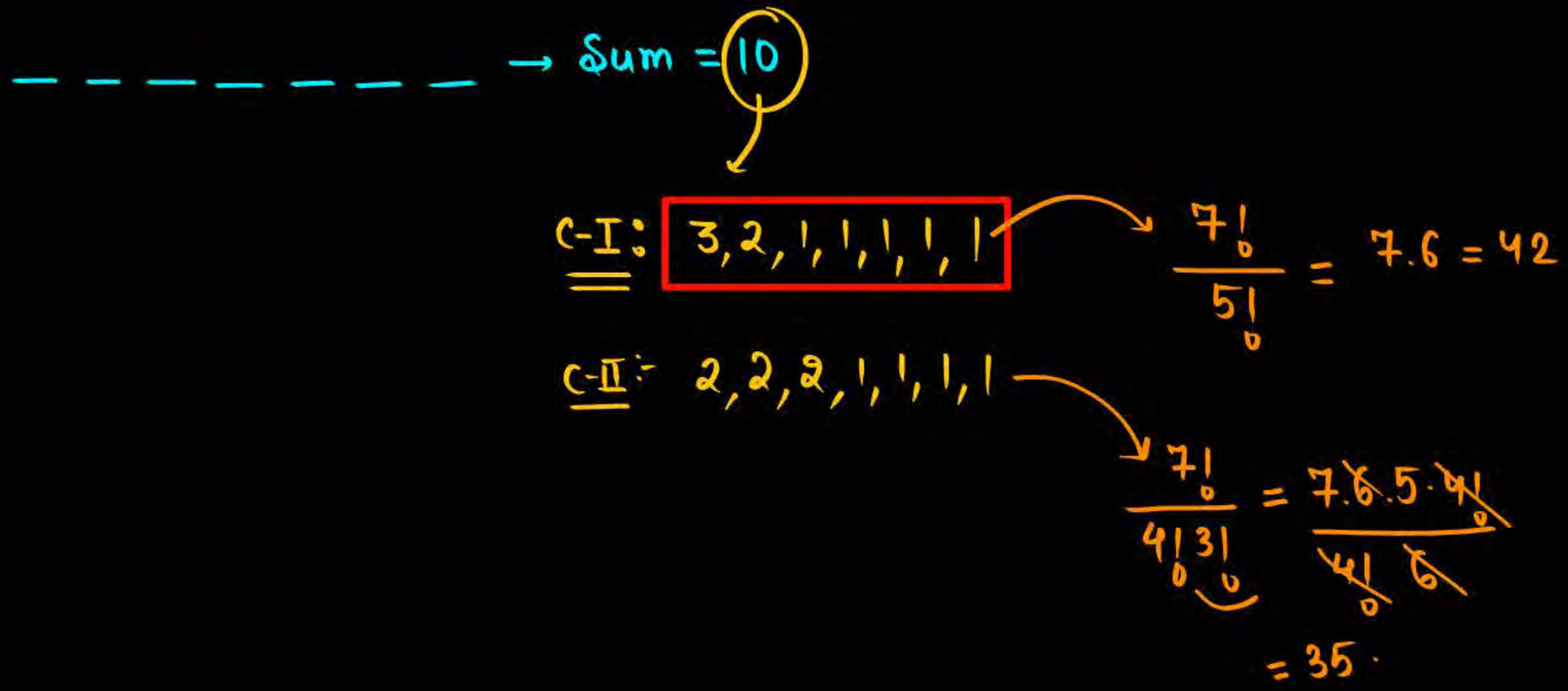
The number of seven digit integers, with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only, is

A 55

B 66

C 77

D 88



## Question [JEE Mains-2025]

[Ans. : 14]

If the number of seven-digit numbers, such that the sum of their digits is even, is $m \cdot n \cdot 10^n; m, n \in \{1, 2, 3, \ldots, 9\}$, then $m + n$ is equal to



# Question Practice

## Question [JEE Mains-2026]

A building has ground floor and 10 more floors. Nine persons enter in a lift at the ground floor. The lift goes up to the $10^{th}$ floor. The number of ways, in which any 4 persons exit at a floor and the remaining 5 persons exit at a different floor, if the does not stop at the first and the second floors, is equal to

HW.

A 2184

B 3064

C 7056

D 11340

## Question [JEE Mains-2023]



Some couples participated in mixed double badminton tournament. If the number of matches played, so that no couple played in a match, is 840, then the total numbers of persons, who participated in the tournament, is _____.



# DO IT BY YOURSELF (DIBY)

# DIBY-15

Eleven books consisting of 5 Mathematics, 4 Physics and 2 Chemistry are placed on a shelf. The number of possible ways of arranging them on the assumption that the books of the same subject are all together is

(A) 4!2! (B) 11! (C) 5!4!3!2! (D) None of these

[Ans. : (C)]

# DIBY-16

The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is

(A) 576 (B) 140 (C) 356 (D) 209

[Ans. : (A)]




$3! \times 5! \times 4! \times 2!$

# DIBY-17

Six married couple are sitting in a room. Then, the number of ways in which 4 people can be selected, so that They do not form a couple, is

(A) 240 (B) 255 (C) 480 (D) 495

[Ans. : (A)]

# DIBY-18

How many of the arrangements of the letter of the word "LOGARITHM" begin with a vowel and end with a consonant?

[Ans. : 90720]

# DIBY-19

Number of permutations of 1, 2, 3, 4, 5, 6, 7, 8 and 9 taken all at a time, such that the Digit 1 appearing somewhere to the left of 2
3 appearing to the left of 4 and
5 somewhere to the left of 6, is
(e.g. 815723946 would be one such permutation)

(A) $9 \cdot 7!$ (B) $8!$ (C) $5! \cdot 4!$ (D) $8! \cdot 4!$

[Ans. : (A)]

# DIBY-20

A test consists of 6 multiple choice questions, each having 4 alternative answers of which only one is correct. The number of ways, in which a candidate answers all six questions such that exactly four of the answers are correct, is

[JEE Mains-2020]

[Ans. : 135]

# DIBY-21

Let the number of elements in sets A and B be five and two respectively. Then the number of subjects of A × B each having at least 3 and at most 6 elements is

[JEE Mains-2023]

(A) 782 (B) 772 (C) 752 (D) 792

[Ans. : (D)]

# DIBY-22

5 Indian & 5 American couples meet at a party & shake hands. If no wife shakes hands with her own husband & no Indian wife shakes hands with a male, then the number of hands shakes that takes place in the party is

(A) 95 (B) 110 (C) 135 (D) 150

[Ans. : (C)]

# DIBY-23

Number of natural numbers between 100 and 1000 such that at least one of their digits is 7, is

(A) 225 (B) 243 (C) 252 (D) none

[Ans. : (C)]

# DIBY-24

Let S = {1, 2, 3, .....9} for $k$ = 1, 2, .....5. Let $N_k$ be the number of subsets of S, each containing five elements out of which exactly $k$ are odd.

Then $N_1 + N_2 + N_3 + N_4 + N_5$ = [JEE (Adv.)-2017]

(A) 125 (B) 210 (C) 252 (D) 126

[Ans. : (.....)]

# DIBY-25

The number of seven-digit numbers, that can be formed by using the digits 1, 2, 3, 5 and 7 such that each digit is used at least once. is: [JEE Mains-2026]

(A) 15400 (B) 17800 (C) 16800 (D) 29400

[Ans. : (C)]

# DIBY-26

Let $p_n$ denote the total number of triangles formed by joining the vertices of an $n$-side regular polygon. If $P_{n+1} - P_n = 66$, then the sum of all distinct prime divisors of $n$ is: [JEE Mains-2026]

(A) 7 (B) 8 (C) 5 (D) 6

[Ans. : (C)]

# DIBY-27

The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is: [JEE Mains-2026]

(A) 5040 (B) 3050 (C) 3410 (D) 4320

[Ans. : (D)]

# DIBY-28

A box contains 5 blue, 6 yellow and 4 red balls The number of ways, of drawing 8 balls containing at least two bails of each colour, is: [JEE Mains-2026]

(A) 4100 (B) 4140 (C) 4230 (D) 4290

[Ans. : (A)]

# DIBY-29

The number of numbers greater than 5000, less than 9000 and divisible by 3, that can be formed using the digits 0, 1, 2, 5, 9, if the repetition of the digits is allowed, is [JEE Mains-2026]

[Ans. : 42]

# DIBY-30

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Let $x$ be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let $y$ be the number of 9-digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,

[JEE Mains-2026]

(A) $56x = 9y$ (B) $29x = 5y$ (C) $45x = 7y$ (D) $21x = 4y$

[Ans. : (D)]

# DIBY-31

There are 12 points in a plane, no three of which are in the same straight line, except 5 points which are collinear. Then the total number of triangles that can be formed with the vertices at any three of these 12 points is

[JEE Mains-2025]

(A) 230 (B) 220 (C) 200 (D) 210

[Ans. : (D)]



# Homework

# Re-attempt all Class Examples

# DPP & DIBY

*It is never about end result*

*It's always about Journey*

# Thank You

#futureIITians